GROWTH OF THE LEGENDRE PARAMETRIX QUOTIENT NEAR THE ENDPOINT PLUS ONE

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Allowed standard prerequisites (editorial).

- The construction of the local parametrix at the endpoint minus one. same article, the lemma at source lines 1363-1390. Provides the pattern that is copied at the endpoint plus one, as the author states at the beginning of the proof.
- The outer parametrix and its matching with the local ones. same article, the display at source line 1351 and the matching condition at source line 1377. Supplies the normalisation with which the constructed local parametrix agrees up to an error of the order of the reciprocal of the parameter.

Source reference. Original equation internal: sources/sigma-2024-004-recurrence-coefficients-for-orthogonal-polyn/pinned-original-source.tex, source lines 1351–1355. The counted statement or proof refers to that the author's displayed outer parametrix; it is the author's own earlier display, definition or statement in the same article and the whole argument of the counted span is retained verbatim.

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properties of the three local parametrices; it is the author's own earlier display, definition or statement in the same article and the whole argument of the counted span is retained verbatim.

Corollary 5.3. *For z in a neighbourhood U_{+1} of $+1$, the matrix-valued function $\widetilde{Q}$ satisfies the asymptotics*

$$\widetilde{Q}(z) = O\big(|z-1|^{-1/4}\big) \tag{5.1}$$

uniformly for $n \to \infty$. Moreover, $\widetilde{Q}$, and its boundary values $\widetilde{Q}_\pm$ on Σ, are bounded in $\overline{\mathbb{C}} := \mathbb{C} \cup \{\infty\}$ away from small neighbourhoods of ± 1, uniformly for $n \to \infty$.

Proof. Analogously to the local parametrix for the Legendre problem at $z = -1$, $\widetilde{P}_{-1}(z) := \widetilde{P}(z)$, one can construct a local parametrix $\widetilde{P}_{+1}(z)$ for the Legendre problem near the point $z = +1$ with local jumps inside an open neighbourhood U_{+1} of $+1$ as depicted in Figure 1.

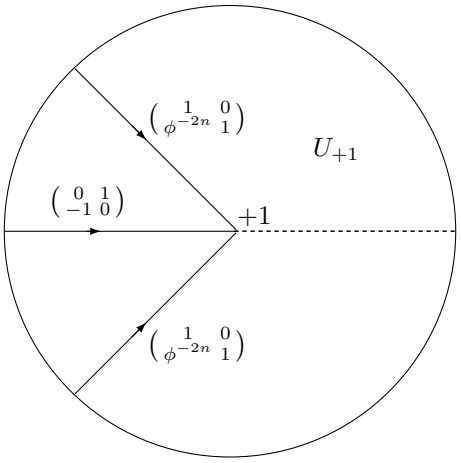


FIGURE 1. The jump conditions for $\widetilde{P}_{+1}$.

In fact, we have $\widetilde{P}_{+1}(z) = \sigma_3 \widetilde{P}_{-1}(-z)\sigma_3$, see [1, equation B.10]. Hence, it follows from Lemma internal (iv) that

$$\widetilde{P}_{+1}(z) = O\big(|z-1|^{-1/4}\big) \tag{5.2}$$

uniformly for $n \to \infty$. After deforming the local contour such that it matches locally with Σ as depicted in Figure 2, we can analytically continue $\widetilde{P}_{+1}$ as necessary to obtain a *deformed* local parametrix $\widetilde{P}^{\mathrm{def}}_{+1}$, which would satisfy locally the same jump conditions as $\widetilde{Q}$.

Because the jump matrices $\widetilde{v}$ and their analytic continuations are uniformly bounded near $z = +1$, $\widetilde{P}^{\mathrm{def}}_{+1}$ would continue to satisfy the estimate (5.2)

$$\widetilde{P}^{\mathrm{def}}_{+1}(z) = O\big(|z-1|^{-1/4}\big) \tag{5.3}$$

uniformly for $n \to \infty$. As the contour deformations are local, we also know that the matching condition (internal) remains unchanged, at least on ∂U_{+1}:

$$\widetilde{Q}(s)\big[\widetilde{P}^{\mathrm{def}}_{+1}(s)\big]^{-1} = I + O\big(n^{-1}\big), \qquad s \in \partial U_{+1}. \tag{5.4}$$

However, as $\widetilde{Q}\big[\widetilde{P}^{\mathrm{def}}_{+1}\big]^{-1}$ has no jumps inside U_{+1}, we can extend (5.4) to all of U_{+1} by the maximum modulus principle for holomorphic functions. Together with (5.3), the estimate (5.1) follows.

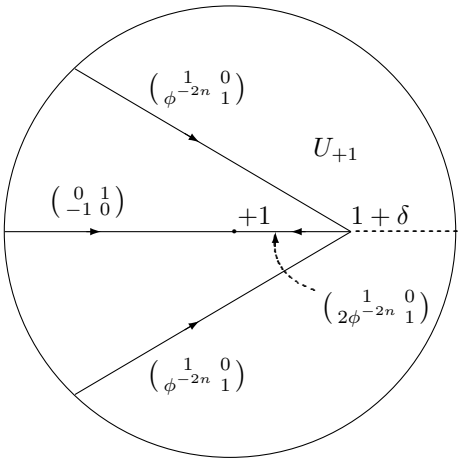


FIGURE 2. The jump conditions for $\widetilde{P}_{+1}^{\mathrm{def}}$.

Regarding the second statement, it has been shown in [2, Section 7] that for a wide class of weight functions including $\widetilde{w}$, the outer parametrix solution N is an approximation to the exact RH solution, in our case $\widetilde{Q}$, uniformly in z as long we stay away from the points ± 1:

$$\big|\widetilde{Q}(z) - N(z)\big| = O\big(n^{-1}\big), \qquad z \text{ staying away from } \pm 1.$$

As can be seen from (internal), N is n-independent and bounded away from the points ± 1. This finishes the proof. □

REFERENCES

[1] Conway T.O., Deift P., Asymptotics of polynomials orthogonal with respect to a logarithmic weight, *SIGMA* **14** (2018), 056, 66 pages, arXiv:1711.01590.

[2] Kuijlaars A.B.J., McLaughlin K.T.-R., Van Assche W., Vanlessen M., The Riemann–Hilbert approach to strong asymptotics for orthogonal polynomials on $[-1, 1]$, *Adv. Math.* **188** (2004), 337–398, arXiv:math.CA/0111252.